

Density of Forces Producing Navier–Stokes Blowup

Shaozhen Cao
shaozhencao@zju.edu.cn

Zhuoni Chi
znchi@zju.edu.cn

Ping Nie
ping.nie@pku.edu.cn

Abstract

We study the density of smooth external forces for which the three-dimensional Navier–Stokes equations lose classical regularity by a prescribed time $T > 0$. Starting from the compact forced blowup solution of OpenAI, we construct a blowup solution near every given smooth solution while preserving its initial velocity. A smooth cutoff of a local vector potential makes the given velocity vanish near the support of a rescaled blowup solution. The two velocities then have no nonlinear interaction, and the resulting force remains smooth through the blowup time. For fixed viscosity and zero initial velocity, such forces are dense in the relative $L_t^1 H_x^s$ topology on both $\mathbb{T}^3$ and $\mathbb{R}^3$ exactly when $s < 1/2$.

Keywords: forced Navier–Stokes equations; finite-time blowup; force-space density; Sobolev spaces; compact support.

1 Introduction

The three-dimensional incompressible Navier–Stokes equations with viscosity $\nu > 0$ and external force f are

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = a. \quad (1)$$

We work on $\mathbb{R}^3$ and on the unit torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. For fixed a and $T > 0$, we ask in which force norms the smooth forces producing classical breakdown by T are dense.

The geometric argument is a divergence-free version of the smooth Urysohn construction: make a given smooth velocity vanish near a small compact set, leave it unchanged outside a slightly larger set, and place a compact blowup solution in the resulting region. Smooth cutoffs are supplied by the classical smooth Urysohn lemma [9, Corollary 1.3.4]. To preserve incompressibility, apply the cutoff to a local vector potential rather than to the velocity itself. The remaining equation error is absorbed into the external force. Our blowup building block is the following result of OpenAI [7, Theorem 1.1].

Theorem 1.1. *For every $\nu > 0$ there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3)$, a compact set $K \subset \mathbb{R}^3$, and smooth velocity and pressure fields u, p on $\mathbb{R}^3 \times [0, 1)$ satisfying*

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = 0, \quad (2)$$

such that $\text{supp } u(\cdot, t) \cup \text{supp } p(\cdot, t) \subset K$ for every $0 \leq t < 1$,

$$\sup_{0 \leq t < 1} \|u(t)\|_{L^2(\mathbb{R}^3)} < \infty, \quad \limsup_{t \uparrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty. \quad (3)$$

Consequently, there is no smooth solution (u, P) on $\mathbb{R}^3 \times [0, \infty)$ with the same force and initial datum whose kinetic energy is uniformly bounded $\sup_{t \geq 0} \frac{1}{2} \int_{\mathbb{R}^3} |u(x, t)|^2 dx < \infty$.

Fix one such building block (U, P, F) [7, 8]. The local gluing strategy parallels Enciso, Peñafiel-Tomás and Peralta-Salas [2, Theorem 1.5, Lemma 2.11, Sections 11.1 and 11.3]: first

modify a given smooth solution locally, then glue in a blowup building block. Their construction produces weak unforced Euler solutions by removing Reynolds stress with convex integration. Here the force is allowed to change, and the glued solution is classical before its blowup time.

More explicitly, for a given smooth velocity $v = \nabla \times A$ locally, choose smooth cutoffs and set

$$w_\varepsilon = -\nabla \times (\eta_\varepsilon \theta_\varepsilon A), \quad u_\varepsilon = v + w_\varepsilon + U_\varepsilon.$$

The corrected velocity $v + w_\varepsilon$ vanishes near the support of the rescaled building block U_ε . Hence both mixed nonlinear terms vanish. The force is the sum of the given force, the rescaled force, and a smooth cutoff correction. Plancherel's theorem and the Fourier dilation formula, followed by the time change $t = t_\varepsilon + \varepsilon^2 \tau$, give the force factor $\varepsilon^{2/q-3/2-s}$ in $L_t^q \dot{H}_x^s$, wherever finite. For $0 \leq s \leq 1$, the corresponding fractional estimate also follows from the classical Sobolev interpolation inequality

$$\|z\|_{\dot{H}^s} \leq \|z\|_2^{1-s} \|\nabla z\|_2^s,$$

which is Hölder's inequality on the Fourier side; see [13, Chapter 4, Sections 1–2]. The cutoff correction has one additional power of ε . Negative orders require the separate low-frequency estimate in Section 4.

The resulting sharp thresholds for zero initial velocity are

Domain	Force topology	Density holds exactly when
$\mathbb{T}^3$	$L_t^1 H_x^s$	$s < 1/2$
$\mathbb{R}^3$	$L_t^1 H_x^s$	$s < 1/2$
$\mathbb{R}^3$	$L_t^2 H_x^s$	$s < -1/2$

Below threshold, density holds for every fixed admissible smooth initial velocity. Near a solution smooth through T , the approximation preserves its initial velocity and earlier history, blows up exactly at T , and converges in energy and dissipation. For a general given force the conclusion is breakdown by T , since a solution that already breaks down earlier needs no modification. Non-density at and above threshold follows from the forced critical regularity estimates, not from scaling alone.

For comparison, Hofmanová, Zhu and Zhu [5, Theorem 1.4, Corollary 4.6, Theorem 5.1] obtain force-density results for nonunique Leray–Hopf solutions. Our property is classical breakdown under a smooth force. Classical local theory and the critical velocity framework are supplied by Tao [11, 12]. Section 2 fixes conventions and collects the standard estimates used below; Sections 3 and 4 give the periodic and whole-space arguments.

2 Definitions and preliminary results

2.1 Norm conventions

We use the unitary angular Fourier transform on $\mathbb{R}^3$ and Fourier series $\hat{z}(k) = \int_{\mathbb{T}^3} z(x) e^{-2\pi i k \cdot x} dx$ on the unit torus. Thus

$$\|z\|_{H^s(\mathbb{R}^3)}^2 = \int (1 + |\xi|^2)^s |\hat{z}(\xi)|^2 d\xi, \quad \|z\|_{H^s(\mathbb{T}^3)}^2 = \sum_k (1 + 4\pi^2 |k|^2)^s |\hat{z}(k)|^2.$$

These are the distributional Sobolev spaces of Taylor [13, Chapters 3–4]; vector and tensor norms sum the squared component norms. The homogeneous norms replace these weights by $|\xi|^{2s}$ and $(2\pi|k|)^{2s}$, respectively, with zero mean on the torus. Tao's unnormalized Fourier convention gives the same Euclidean norms after its Plancherel factor [10, Appendix A].

Time norms are Bochner norms of strongly measurable paths, with equality almost everywhere understood [6, Definitions 6.14, 6.27]:

$$\|z\|_{L^q(I;X)} = \left(\int_I \|z(t)\|_X^q dt \right)^{1/q}, \quad \|z\|_{L^\infty(I;X)} = \operatorname{ess\,sup}_{t \in I} \|z(t)\|_X. \quad (4)$$

Force norms use $(0, \infty)$ unless stated otherwise, and give each smooth force class its relative norm topology. On $(0, T)$ we write

$$\|z\|_{E_T} = \|z\|_{L_t^\infty L_x^2} + \|\nabla z\|_{L_t^2 L_x^2}. \quad (5)$$

This is equivalent to the sum norm on $L_t^\infty L_x^2 \cap L_t^2 H_x^1$; it assigns no value at the terminal time.

2.2 Smooth data and solution classes

All fields are real, and forces need not be divergence free. We write $L_\sigma^2(D)$ for the divergence-free subspace of $L^2(D; \mathbb{R}^3)$. The allowed initial velocities and forces are

$$\mathcal{X} = \mathcal{X}_{\mathbb{T}^3} = C_{\operatorname{div}}^\infty(\mathbb{T}^3; \mathbb{R}^3), \quad \mathcal{F} = \mathcal{F}_{\mathbb{T}^3} = C_c^\infty(\mathbb{T}^3 \times (0, \infty); \mathbb{R}^3), \quad (6)$$

$$\mathcal{X}_{\mathbb{R}} = H^\infty(\mathbb{R}^3; \mathbb{R}^3) \cap L_\sigma^2(\mathbb{R}^3), \quad (7)$$

where $H^\infty = \bigcap_{m=0}^\infty H^m$ has its usual Fréchet topology. On the whole space set

$$\mathcal{F}_{\mathbb{R}} = \left\{ f \in C^\infty([0, \infty); H^\infty) : \|f\|_{L_t^1 H_x^m} + \|f\|_{L_t^2 H_x^m} < \infty \text{ for every integer } m \geq 0 \right\}. \quad (8)$$

Smoothness into H^∞ means smoothness into each H^m , with one-sided time derivatives at zero. Thus periodic forces vanish near zero and outside a compact time interval; whole-space forces may be nonzero at zero and need not have compact support. Both classes consist of forces defined through any possible blowup time of the velocity.

On the torus, velocity and pressure are periodic and $\int_{\mathbb{T}^3} p = 0$. On $\mathbb{R}^3$, a classical velocity belongs to $C([0, S]; H^m)$ for every integer $m \geq 0$ on each compact interval of its lifespan. Its pressure gradient is specified below, and the scalar pressure is determined up to a function of time. In either case local uniqueness defines a maximal classical lifespan, denoted by $T_{\max}^\nu(a, f)$ on the torus and $T_{\max, \mathbb{R}}^\nu(a, f)$ on the whole space. A given smooth solution is *smooth through* T if it extends smoothly to $[0, T + \delta]$ for some $\delta > 0$ in the relevant class.

For fixed $\nu, T > 0$, define

$$\mathcal{B}_{\nu, a, T} = \{f \in \mathcal{F} : T_{\max}^\nu(a, f) \leq T\}, \quad \mathcal{B}_{\nu, T}^0 = \mathcal{B}_{\nu, 0, T}, \quad (9)$$

$$\mathcal{B}_{\nu, a, T}^{\mathbb{R}} = \{f \in \mathcal{F}_{\mathbb{R}} : T_{\max, \mathbb{R}}^\nu(a, f) \leq T\}. \quad (10)$$

We call these the forces producing classical breakdown by T , using “breakdown for failure of classical continuation, as in the Clay problem statement [3, alternatives (C) and (D)]. The glued solutions satisfy the more specific condition of unbounded speed at their terminal time.

2.3 Homogeneous spaces and Fourier multipliers

Write $\Lambda = (-\Delta)^{1/2}$ and $J = (I - \Delta)^{1/2}$, with the Fourier conventions of Section 1. On mean-zero periodic fields, the homogeneous and inhomogeneous Sobolev norms are equivalent at each fixed order. For $0 < a < 3/2$, the homogeneous completion is represented by $\widehat{z} = |\xi|^{-a} G$, $G \in L^2$, with its unique $L^{6/(3-2a)}$ representative supplied by Sobolev embedding. The distributional pairing is well defined since $|\xi|^{-a}$ is locally square integrable and Schwartz tests decay rapidly. For the completed energy-force space we fix

$$\dot{H}^{-1}(\mathbb{R}^3) = \{h \in \mathcal{S}' : \widehat{h} = F \text{ is a measurable function and } |\xi|^{-1} F \in L^2\}. \quad (11)$$

Functions representing distributions are identified almost everywhere. The maps $h \mapsto |\xi|^{-1}\widehat{h}$ and $h \mapsto \langle \xi \rangle^s \widehat{h}$ identify these spaces with L^2 ; the inverse Fourier data define tempered distributions. Thus they are separable Hilbert spaces, with no polynomial ambiguity. Real vector fields correspond to the conjugate-symmetric Fourier data.

2.4 Standard Sobolev estimates

We use the tame product estimate and $H^2 \hookrightarrow L^\infty$:

$$\|vw\|_{H^m} \leq C_m(\|v\|_{H^2} \|w\|_{H^m} + \|w\|_{H^2} \|v\|_{H^m}), \quad \|v\|_\infty \leq C \|v\|_{H^2}, \quad m \geq 2. \quad (12)$$

See Tao [10, Appendix A, (A.13), Lemma A.8]; the periodic version follows by the same Fourier convolution estimate, with $\|\widehat{v}\|_{\ell^1} \leq C\|v\|_{H^2}$. The critical embeddings used below are

$$\|v\|_3 \leq C \|v\|_{\dot{H}^{1/2}}, \quad \|\nabla v\|_3 + \|\Lambda v\|_3 \leq C \|v\|_{\dot{H}^{3/2}}, \quad \|\nabla v\|_6 \leq C \|\Delta v\|_2. \quad (13)$$

They follow from the homogeneous Sobolev inequality [10, Appendix A, (A.11)], applied also to derivatives. On the torus use the compact-manifold embedding [14, Chapter 13, Proposition 6.4 and preceding discussion] and the spectral gap for mean-zero fields. These statements use the homogeneous representatives fixed above; no embedding of $\dot{H}^{3/2}$ into L^∞ is used.

2.5 Projection, pressure, and local existence

The Leray projection $\mathbb{P}$ has Fourier symbol $I - \xi \otimes \xi / |\xi|^2$ on $\mathbb{R}^3 \setminus \{0\}$ and the same symbol at $k \neq 0$ on the torus. At the periodic zero mode it is the identity. It is bounded on every H^s and commutes with derivatives and the heat semigroup. The projected equation is

$$\partial_t u - \nu \Delta u = -\mathbb{P} \nabla \cdot (u \otimes u) + \mathbb{P} f. \quad (14)$$

On the torus, pressure is recovered from

$$\Delta p = \nabla \cdot f - \sum_{i,j} \partial_i \partial_j (u_i u_j), \quad \int_{\mathbb{T}^3} p = 0.$$

The right side has zero mean. On $\mathbb{R}^3$ we require

$$\nabla p = (I - \mathbb{P})(f - \nabla \cdot (u \otimes u)) =: G. \quad (15)$$

The field G is curl free. The radial potential $p(x, t) = \int_0^1 G(rx, t) \cdot x \, dr$ therefore recovers the pressure, up to a function of time. We impose no L^2 condition on p itself; $\nabla p \in L^2$ excludes nonzero constant pressure gradients.

Proposition 2.1. *For each initial velocity in the stated class and each force smooth into every H^m on compact time intervals, equation (1) has a unique maximal smooth velocity, with pressure determined as above. If a solution is defined on $[0, S)$, $S < \infty$, and*

$$\int_0^S \|u(t)\|_{H^2(D)}^2 \, dt < \infty, \quad (16)$$

then it extends smoothly beyond S .

Proof. Apply Tao [11, Theorems 5.1(ii)–(iv), 5.4(ii)–(iv)] to $\mathbb{P}f$. The proof of Theorem 5.4(iv), p. 53, explicitly permits $a \in H^k$ and $f \in C_t^j H_x^k$ for all j, k ; the resulting smoothness is on one common interval. Reduce viscosity to one by $\tau = \nu t$, $\tilde{u} = \nu^{-1}u$, and $(\tilde{p}, \tilde{f}) = \nu^{-2}(p, f)$ with the corresponding time change. For nonzero periodic mean, use Tao's mean reduction [11, Section 3, (36)–(38)]: set $m(t) = \int_{\mathbb{T}^3} a + \int_0^t \int_{\mathbb{T}^3} f$ and translate by $X(t) = \int_0^t m$, subtracting

m from the velocity and m' from the force. Pressure is recovered as above. Uniqueness and maximal development follow from the same theorems and the overlap argument of Corollary 5.2.

For continuation, the standard H^m energy estimate, Young's inequality and (12) give, for $m \geq 3$,

$$(\|u\|_{H^m})' \leq C_{m,\nu} \|u\|_{H^2}^2 \|u\|_{H^m} + \|f\|_{H^m}. \quad (17)$$

Regularize the norm at zero. Grönwall and (16) bound every H^m norm uniformly up to S . Tao's quantitative local existence then gives a uniform positive time when restarting at $t_0 \uparrow S$; the fixed force is smooth on $[0, S+1]$. Uniqueness glues one such solution past S . $\square$

2.6 Energy and initial vanishing of the compact solution

We use the fixed solution (U, P, F) chosen after Theorem 1.1. Its energy estimate supplies the time-integrated dissipation bound and the initial vanishing needed for rescaling.

Lemma 2.2. *The building block satisfies*

$$M := \sup_{0 \leq t < 1} \|U(t)\|_2 < \infty, \quad D := \|\nabla U\|_{L^2((0,1) \times \mathbb{R}^3)} < \infty.$$

Both U and P vanish on an initial time interval and therefore extend smoothly by zero to negative times.

Proof. The classical energy identity and $U(0) = 0$ give, with $N(t) = \int_0^t \|F(s)\|_2 ds$, the bounds $\|U(t)\|_2 \leq N(t)$ and

$$\|U(t)\|_2^2 + 2\nu \int_0^t \|\nabla U(s)\|_2^2 ds \leq 2 \int_0^t \|F(s)\|_2 N(s) ds = N(t)^2. \quad (18)$$

Compact support justifies the identity; regularizing $\|U\|_2$ justifies the first bound at zero. Since F is time compact, monotone convergence gives finite dissipation up to time one. Before the support of F , the same estimate gives $U = 0$; the equation and compact support then give $P = 0$. Both fields therefore extend smoothly by zero to negative times. $\square$

3 The torus

All spatial norms in this section are on $\mathbb{T}^3$, unless another domain is shown. The spaces $\mathcal{X}$, $\mathcal{F}$, and $\mathcal{B}_{\nu,a,T}$ are those defined in Section 2.

3.1 Statement of the density theorem

Theorem 3.1 (Sobolev density threshold). *Equip $\mathcal{F}$ with its relative $L^1(0, \infty; H^s(\mathbb{T}^3))$ norm topology.*

- (i) *For every fixed $a \in \mathcal{X}$, the set $\mathcal{B}_{\nu,a,T}$ is dense if $s < 1/2$.*
- (ii) *For zero initial velocity,*

$$\mathcal{B}_{\nu,T}^0 \text{ is dense in } \mathcal{F} \iff s < \frac{1}{2}.$$

We first construct a blowup solution close to a given smooth solution. The force estimates yield density below the stated threshold. A separate small-force estimate then proves non-density at and above it.

3.2 Localization and scaling

Lemma 3.2. *Let B be a fixed coordinate ball whose closure lies inside a torus fundamental cube. If z is smooth and supported in B , let $z_{\mathbb{R}}$ be its zero extension in that cube to $\mathbb{R}^3$, and let $z_{\mathbb{T}^3}$ be its periodization. For $0 < s < 1$,*

$$\|z_{\mathbb{T}^3}\|_{H^s(\mathbb{T}^3)} \leq C_{s,B}(\|z_{\mathbb{R}}\|_2 + \|z_{\mathbb{R}}\|_{\dot{H}^s(\mathbb{R}^3)}). \quad (19)$$

The constant is uniform as the support shrinks inside B . At $s = 0$ the L^2 norms are equal; at $s = 1$ the corresponding equality holds for the L^2 gradient norms.

Proof. Sobolev norms on a compact manifold are equivalent to the norms obtained by a fixed coordinate cover and partition of unity [13, Chapter 4, Sections 2–3]. Choose one fixed cutoff equal to one near $\bar{B}$ and supported in the coordinate chart. The resulting local-to-global map is bounded from $H^s(\mathbb{R}^3)$ to $H^s(\mathbb{T}^3)$. Since $\|z\|_{H^s(\mathbb{R}^3)} \leq C_s(\|z\|_2 + \|z\|_{\dot{H}^s})$, this gives (19). The chart and cutoff are fixed, so the constant is independent of a smaller support of z . At orders zero and one, integration over the single copy of the support gives the stated identities. $\square$

Choose a compact set K_* containing K and the spatial projection of $\text{supp } F$. Fix $x_0 \in B$, where B is the ball in Lemma 3.2. For sufficiently small $\varepsilon > 0$, require

$$2\varepsilon^2 < T, \quad x_0 + \varepsilon K_* \subset B, \quad t_\varepsilon = T - \varepsilon^2.$$

We extend F by zero to nonpositive source times; this extension is smooth because its temporal support is compact in $(0, \infty)$. Using this extension and the negative-time zero extensions of U, P from Lemma 2.2, define

$$\begin{aligned} U_\varepsilon(x, t) &= \varepsilon^{-1} U\left(\frac{x - x_0}{\varepsilon}, \frac{t - t_\varepsilon}{\varepsilon^2}\right), \\ P_\varepsilon(x, t) &= \varepsilon^{-2} P\left(\frac{x - x_0}{\varepsilon}, \frac{t - t_\varepsilon}{\varepsilon^2}\right), \\ F_\varepsilon(x, t) &= \varepsilon^{-3} F\left(\frac{x - x_0}{\varepsilon}, \frac{t - t_\varepsilon}{\varepsilon^2}\right). \end{aligned} \quad (20)$$

The first two fields are used for $t < T$; the force is defined for all positive times. Place the rescaled whole-space solution in B and then periodize it. The torus contains a single copy of its support.

Proposition 3.3. *The fields in (20) solve the periodic momentum equation at viscosity ν , start from zero, and have unbounded speed at T . A spatially constant pressure adjustment enforces the mean-zero normalization. Their norms satisfy*

$$\|U_\varepsilon\|_{L^\infty(0, T; L^2)} = \varepsilon^{1/2} M, \quad \|\nabla U_\varepsilon\|_{L^2(0, T; L^2)} = \varepsilon^{1/2} D, \quad (21)$$

$$\|F_\varepsilon\|_{L^q(0, \infty; L^p)} = \varepsilon^{\alpha(p, q)} \|F\|_{L^q(0, \infty; L^p)}, \quad \alpha(p, q) = -3 + \frac{3}{p} + \frac{2}{q}, \quad (22)$$

for $1 \leq p, q \leq \infty$. For $0 \leq s \leq 1$,

$$\|F_\varepsilon\|_{L^1(0, \infty; H^s(\mathbb{T}^3))} \leq C_s(\varepsilon^{1/2} + \varepsilon^{1/2-s}). \quad (23)$$

In particular the force tends to zero in $L_t^1 H_x^s$ for every $s < 1/2$, including negative s .

Proof. Writing $y = (x - x_0)/\varepsilon$ and $\sigma = (t - t_\varepsilon)/\varepsilon^2$, every term of the momentum equation gains the common factor ε^{-3} . Incompressibility is preserved and no viscosity rescaling occurs. The temporal extension is smooth by initial vanishing. The speed identity $\|U_\varepsilon(t)\|_\infty = \varepsilon^{-1} \|U(\sigma)\|_\infty$ proves unboundedness as $t \uparrow T$.

At a fixed time the velocity has amplitude factor ε^{-1} and spatial volume factor ε^3 , giving the first identity in (21). Its gradient has amplitude factor ε^{-2} . Squaring and integrating in both space and time gives $\varepsilon^{-4}\varepsilon^3\varepsilon^2 = \varepsilon$ times the original dissipation integral, proving the second identity.

For finite p, q , the force's spatial norm gains $\varepsilon^{-3+3/p}$ and its time norm gains $\varepsilon^{2/q}$. The positive-time domain includes the whole transformed support because $t_\varepsilon > 0$ and the original force vanishes at negative times. This proves (22); the essential-supremum cases follow by the same change of variables with zero reciprocal exponents.

Euclidean homogeneous Sobolev scaling gives, for $0 < s < 1$,

$$\left\| \varepsilon^{-3} F((\cdot - x_0)/\varepsilon, \sigma) \right\|_{\dot{H}^s(\mathbb{R}^3)} = \varepsilon^{-3/2-s} \|F(\cdot, \sigma)\|_{\dot{H}^s(\mathbb{R}^3)}.$$

The Fourier transform gives the squared homogeneous scaling factor $\varepsilon^{-6}\varepsilon^6\varepsilon^{-3-2s} = \varepsilon^{-3-2s}$. Time integration contributes ε^2 . The inhomogeneous L^2 term has order $\varepsilon^{1/2}$. Lemma 3.2 therefore proves (23). At $s = 0$ use the mixed-norm identity, and at $s = 1$ use the gradient scaling. For $s < 0$, the Fourier definition gives $\|z\|_{H^s} \leq \|z\|_2$, so the $s = 0$ estimate proves convergence without any homogeneous negative-order claim. $\square$

The force amplitude is $\varepsilon^{-3} \|F\|_\infty$, even though its integrated norm tends to zero below the critical order. The homogeneous $L_t^1 \dot{H}_x^{1/2}$ norm is scale invariant. Section 3.5 supplies the regularity estimate needed at that order.

3.3 Gluing to a given smooth solution

Fix a given smooth solution (v, π, g) smooth on $[0, T + \delta]$, where $\delta > 0$, $g \in \mathcal{F}$, and $v(0) = a \in \mathcal{X}$. We first make the given velocity vanish near the support of U_ε . This prevents nonlinear interaction terms from introducing a singularity into the new force.

Choose $R > 0$ with $K_* \subset B(0, R)$. The smooth Urysohn lemma [9, Corollary 1.3.4] gives an open neighborhood V of K_* and smooth cutoff functions satisfying

$$\begin{aligned} \theta &\in C_c^\infty(\mathbb{R}^3), \quad 0 \leq \theta \leq 1, \quad \theta|_V = 1, \quad \text{supp } \theta \subset B(0, R), \\ \eta &\in C_c^\infty(\mathbb{R}), \quad 0 \leq \eta \leq 1, \quad \eta|_{[-1,1]} = 1, \quad \text{supp } \eta \subset (-2, 2). \end{aligned}$$

These choices are fixed independently of ε ; their rescalings below localize the modification near x_0 in space and near T in time.

Lemma 3.4. *Let v be smooth and divergence free in a spatial ball centered at x_0 . In that ball, with $y = x - x_0$, define*

$$A(x, t) = \int_0^1 r v(x_0 + ry, t) \times y \, dr. \quad (24)$$

Then $\nabla \times A = v$. With the fixed cutoffs θ and η above, put

$$\theta_\varepsilon(x) = \theta((x - x_0)/\varepsilon), \quad \eta_\varepsilon(t) = \eta((t - T)/\varepsilon^2), \quad w_\varepsilon = -\nabla \times (\eta_\varepsilon \theta_\varepsilon A). \quad (25)$$

For sufficiently small ε , w_ε is a smooth, divergence-free field supported inside the coordinate ball and in $(T - 2\varepsilon^2, T + 2\varepsilon^2)$. During the active interval before blowup of U_ε ,

$$v + w_\varepsilon = 0 \quad \text{on an open neighborhood of } \text{supp } U_\varepsilon(t). \quad (26)$$

Proof. Differentiating the radial formula and using $\nabla \cdot v = 0$ gives $\nabla \times A = v$. Hence w_ε is divergence free, and on the common plateau of the cutoffs it equals $-v$. For sufficiently small ε the spatial support lies strictly inside the coordinate ball and the temporal support lies in $(0, T + \delta)$, so extension by zero is smooth. The rescaled building block is active only for $T - \varepsilon^2 \leq t < T$, where $\eta_\varepsilon = 1$; its support lies in the spatial plateau. This proves (26) and the support assertions. $\square$

Lemma 3.5. For w_ε in Lemma 3.4, define

$$H_\varepsilon = \partial_t w_\varepsilon - \nu \Delta w_\varepsilon + (v \cdot \nabla) w_\varepsilon + (w_\varepsilon \cdot \nabla) v + (w_\varepsilon \cdot \nabla) w_\varepsilon. \quad (27)$$

The force H_ε is smooth across T and extends by zero to all times outside its cutoff support. Its spatial support has volume $O(\varepsilon^3)$, and its temporal support has length $O(\varepsilon^2)$. For every spatial multi-index β and integer $j \geq 0$,

$$|\partial_t^j \partial_x^\beta w_\varepsilon| \leq C_{\beta,j} \varepsilon^{-2j-|\beta|}, \quad |\partial_x^\beta H_\varepsilon| \leq C_\beta \varepsilon^{-2-|\beta|}. \quad (28)$$

In particular,

$$\|w_\varepsilon\|_{E_T} \leq C \varepsilon^{3/2}, \quad (29)$$

$$\|H_\varepsilon\|_{L^q(0,\infty;L^p)} \leq C_{p,q} \varepsilon^{-2+3/p+2/q} = C_{p,q} \varepsilon^{\alpha(p,q)+1}, \quad (30)$$

$$\|H_\varepsilon\|_{L^1(0,\infty;H^s)} \leq C_s (\varepsilon^{3/2} + \varepsilon^{3/2-s}) \quad (0 \leq s \leq 1). \quad (31)$$

All constants are independent of sufficiently small ε .

Proof. Set $x = x_0 + \varepsilon z$, $t = T + \varepsilon^2 \sigma$. The radial potential has $A(x, t) = \varepsilon \mathcal{A}_\varepsilon(z, \sigma)$, where $\mathcal{A}_\varepsilon$ and all its derivatives are uniformly bounded on the fixed cutoff support. Thus

$$w_\varepsilon(x, t) = W_\varepsilon(z, \sigma), \quad W_\varepsilon = -\nabla_z \times (\eta(\sigma) \theta(z) \mathcal{A}_\varepsilon).$$

Substitution in (27) gives $H_\varepsilon(x, t) = \varepsilon^{-2} Q_\varepsilon(z, \sigma)$, where Q_ε is uniformly smooth and supported in the same fixed cylinder: its terms are $\partial_\sigma W_\varepsilon$, $-\nu \Delta_z W_\varepsilon$ and transport terms with nonnegative powers of ε and bounded coefficients. This proves (28). Changing variables gives (29)–(30); applying Sobolev scaling to Q_ε and then Lemma 3.2 gives (31). The smooth cutoffs give the asserted zero extensions in time and space. $\square$

Theorem 3.6. Let (v, π, g) be a periodic solution smooth on $[0, T + \delta]$, with $\delta > 0$, $g \in \mathcal{F}$ and $v(0) = a \in \mathcal{X}$. Fix any nonempty coordinate ball. For all sufficiently small $\varepsilon > 0$, there are $g_\varepsilon \in \mathcal{F}$ and a solution u_ε with

- (i) $T_{\max}^\nu(a, g_\varepsilon) = T$ and $\limsup_{t \uparrow T} \|u_\varepsilon(t)\|_\infty = \infty$;
- (ii) $u_\varepsilon = v$ for $0 \leq t \leq T - 2\varepsilon^2$;
- (iii) $u_\varepsilon - v$ divergence free and supported, for every $t < T$, in a ball of diameter $O(\varepsilon)$ inside the chosen ball;
- (iv) the simultaneous bounds

$$\|u_\varepsilon - v\|_{E_T} \leq (M + D) \varepsilon^{1/2} + C \varepsilon^{3/2}, \quad (32)$$

$$\|g_\varepsilon - g\|_{L_t^q L_x^p} \leq C_{p,q} (\varepsilon^{\alpha(p,q)} + \varepsilon^{\alpha(p,q)+1}), \quad (33)$$

$$\|g_\varepsilon - g\|_{L_t^1 H_x^s} \leq C_s (\varepsilon^{1/2-s} + \varepsilon^{3/2-s}) \quad (0 \leq s < 1/2). \quad (34)$$

Here force norms are taken over $(0, \infty)$, and $\alpha(p, q) = -3 + 3/p + 2/q$ for $1 \leq p, q \leq \infty$. For $s < 0$, the force difference also tends to zero in $L_t^1 H_x^s$.

Proof. Choose the rescaled blowup solution and the correction from Lemmas 3.4–3.5, with all supports inside the prescribed ball, and set

$$u_\varepsilon = v + w_\varepsilon + U_\varepsilon, \quad p_\varepsilon = \pi + P_\varepsilon, \quad g_\varepsilon = g + H_\varepsilon + F_\varepsilon. \quad (35)$$

Initially use the compact representative P_ε ; subtract its spatial mean to normalize p_ε .

Let $b_\varepsilon = v + w_\varepsilon$. Expanding the equation for the given smooth solution and (27) gives

$$\partial_t b_\varepsilon + (b_\varepsilon \cdot \nabla) b_\varepsilon - \nu \Delta b_\varepsilon + \nabla \pi = g + H_\varepsilon.$$

Adding the building block equation produces the equation for u_ε except for

$$(b_\varepsilon \cdot \nabla)U_\varepsilon + (U_\varepsilon \cdot \nabla)b_\varepsilon.$$

Both terms are identically zero. On a neighborhood of $\text{supp } U_\varepsilon$ this follows from $b_\varepsilon = 0$, including its derivatives, by (26). Outside that support U_ε and its derivatives vanish, including at the support boundary by smoothness. Before its starting time it vanishes everywhere. Thus (1) holds exactly for (35) on $[0, T)$.

Each summand of the velocity is divergence free. The correction is zero for $t \leq T - 2\varepsilon^2$, and the building block is zero before $T - \varepsilon^2$, so the initial datum and the stated earlier history are unchanged. On the active building block support, $u_\varepsilon = U_\varepsilon$. Hence

$$\|u_\varepsilon(t)\|_\infty \geq \|U_\varepsilon(t)\|_\infty$$

and the maximum norm is unbounded as $t \uparrow T$.

The velocity is smooth on every closed interval before blowup. The force F_ε is globally smooth by its original definition, and H_ε is globally smooth by Lemma 3.5. Their time supports remain compact subsets of $(0, \infty)$ when ε is sufficiently small. Therefore $g_\varepsilon \in \mathcal{F}$, independently of the absence of a classical extension of u_ε at T . Uniqueness in Proposition 2.1 identifies the constructed velocity with the maximal solution for (a, g_ε) up to every $T' < T$. Its lifespan is at least T , and the unbounded maximum norm excludes extension beyond T . Thus the lifespan is exactly T .

Finally, the triangle inequality, Proposition 3.3 and Lemma 3.5 give (32)–(34). In (34), the lower-order inhomogeneous terms are absorbed because $0 < \varepsilon \leq 1$ and $s \geq 0$. For negative s , use $\|z\|_{H^s} \leq \|z\|_2$ and the $s = 0$ estimate. $\square$

3.4 Density below the critical order

Proposition 3.7. *For $a \in \mathcal{X}$, $\nu > 0$, $T > 0$ and $s < 1/2$,*

$$\forall g \in \mathcal{F} \ \forall \rho > 0 \ \exists f \in \mathcal{F} : \quad \|f - g\|_{L_t^1 H_x^s} < \rho, \quad T_{\max}^\nu(a, f) \leq T.$$

Proof. Fix g and ρ . There are two exhaustive cases.

If $T_{\max}^\nu(a, g) \leq T$, choose $f = g$. It already belongs to $\mathcal{B}_{\nu, a, T}$ and the norm difference is zero.

If $T_{\max}^\nu(a, g) > T$, choose $\delta > 0$ so that $T + \delta < T_{\max}^\nu(a, g)$, and use its smooth solution as the given smooth solution in Theorem 3.6. For every sufficiently small ε , that theorem supplies $g_\varepsilon \in \mathcal{B}_{\nu, a, T}$ with lifespan exactly T . For $0 \leq s < 1/2$, (34) tends to zero. For $s < 0$, its $s = 0$ case and the embedding $L^2 \hookrightarrow H^s$ do so. Choose ε so that the force difference is less than ρ and set $f = g_\varepsilon$.

Both choices preserve a , ν , and T , and give the required approximation. $\square$

3.5 Critical regularity and completion of the proof

Non-density at and above the threshold is a consequence of classical stability of the zero solution in the critical topology. For a direct periodic robustness theorem with $L_t^2 H_x^{-1/2}$ force perturbations, see Marín-Rubio, Robinson and Sadowski [16, Theorem 3, author manuscript pp. 6–7]. The $L_t^1 H_x^{1/2}$ small-force result needed here follows from the forced Fujita–Kato theory in Danchin [15, Theorem 2.3.1, pp. 47–49]; we record only the adaptation of its hypotheses.

Proposition 3.8. *There is $c > 0$, depending only on the unit torus and the stated norm conventions, such that*

$$g \in \mathcal{F}, \quad \rho := \|g\|_{L^1(0, \infty; H^{1/2})} < c\nu \tag{36}$$

imply $T_{\max}^\nu(0, g) = \infty$.

Proof. First remove the mean as in Tao [11, Section 3, (36)–(38)]. Set $m(t) = \int_0^t \bar{g}(\tau) d\tau$ and $b(t) = \int_0^t m(\tau) d\tau$. In the translated coordinates, $v(t, x) = u(t, x + b(t)) - m(t)$ solves the mean-zero periodic equation with force $h(t, x) = g(t, x + b(t)) - \bar{g}(t)$. Translation is an isometry and removal of the zero mode decreases the norm, so $\|h\|_{L_t^1 H_x^{1/2}} \leq \rho$ and $|m(t)| \leq \rho$.

Apply the critical small-data fixed-point theorem [15, Theorem 2.3.1 and its proof] with $p = r = 2$. Here $\dot{B}_{2,2}^{1/2} = \dot{H}^{1/2}$, and Minkowski's inequality gives $\|h\|_{\tilde{L}^1 \dot{H}^{1/2}} \leq \|h\|_{L^1 \dot{H}^{1/2}}$. The same proof applies to mean-zero periodic fields using the periodic heat estimates in [15, Section 2.2.4, p. 44]; the zero mode has already been removed. It gives a global critical solution when $\rho < c\nu$. Smooth data retain their higher regularity by the local theory cited in Proposition 2.1. Restoring the translation and bounded mean therefore gives the required global classical solution. $\square$

Corollary 3.9. *For every $s \geq 1/2$ and $T > 0$, $\mathcal{B}_{\nu,T}^0$ is not dense in $\mathcal{F}$ with the relative $L_t^1 H_x^s$ topology.*

Proof. At $s = 1/2$, the set

$$\{g \in \mathcal{F} : \|g\|_{L_t^1 H_x^{1/2}} < c\nu\}$$

is a nonempty relative open ball, containing zero, and is disjoint from $\mathcal{B}_{\nu,T}^0$ by Proposition 3.8. For $s \geq 1/2$, the Fourier weights imply $\|g(t)\|_{H^{1/2}} \leq \|g(t)\|_{H^s}$. The ball $\|g\|_{L_t^1 H_x^s} < c\nu$ is therefore also disjoint from the set of forces producing breakdown. A dense subset cannot miss a nonempty open set. $\square$

Proof of Theorem 3.1. Proposition 3.7 gives density for every fixed smooth initial velocity when $s < 1/2$. Corollary 3.9 gives non-density for zero initial velocity when $s \geq 1/2$. These are the two assertions of the theorem. $\square$

The converse has been proved for zero initial velocity. The behavior for general nonzero initial velocities at or above the critical order remains outside this classification.

3.6 Other force norms, trajectory approximation, and data pairs

Corollary 3.10. *For every fixed $a \in \mathcal{X}$, the set $\mathcal{B}_{\nu,a,T}$ is dense in $\mathcal{F}$ with its relative $L^q(0, \infty; L^p)$ topology whenever*

$$1 \leq p, q \leq \infty, \quad \frac{3}{p} + \frac{2}{q} > 3. \quad (37)$$

Proof. Use the same two cases as in Proposition 3.7. In the case of a solution smooth through the prescribed time, (33) tends to zero because $\alpha(p, q) > 0$ and hence $\alpha(p, q) + 1 > 0$. $\square$

The sufficient region includes $L_t^1 L_x^2$ and $L_t^2 L_x^{4/3}$. The other mixed Lebesgue topologies require further analysis.

Corollary 3.11. *Fix $a \in \mathcal{X}$. Let $\mathcal{R}_{a,T}$ be the trajectories with initial velocity a and forces in $\mathcal{F}$ that are smooth through T . Let $\mathcal{S}_{a,T}$ be the trajectories with the same initial velocity and forces in $\mathcal{F}$ that are smooth on $[0, T)$, have finite E_T norm, and have unbounded maximum velocity at T . Then*

$$\mathcal{R}_{a,T} \subseteq \overline{\mathcal{S}_{a,T}}^{E_T}. \quad (38)$$

For each given pair (v, g) and every fixed $s < 1/2$, the approximating pairs can be chosen with

$$(u_\varepsilon, g_\varepsilon) \longrightarrow (v, g) \quad \text{in } E_T \times L^1(0, \infty; H^s).$$

Proof. A given solution smooth through T is smooth through $T + \delta$ for some $\delta > 0$, by local continuation; equivalently one may take this as the meaning of smoothness through the endpoint. Apply Theorem 3.6. Its energy estimate gives convergence in E_T , and its force estimates give simultaneous convergence. The given smooth solution has finite E_T norm and the building block and correction do also, so u_ε belongs to the specified ambient energy space.

To verify the claimed interpretation of the norm, note that

$$\|z\|_{L^2(0,T;L^2)} \leq T^{1/2} \|z\|_{L^\infty(0,T;L^2)}.$$

Thus E_T controls both terms of $L_t^2 H_x^1$, while the usual sum norm controls E_T directly. This proves the equivalence used in (38). No endpoint value or classical continuation of u_ε is implied by convergence in this norm. $\square$

Proposition 3.12. *Define*

$$\mathfrak{B}_{\nu,T} = \{(a, f) \in \mathcal{X} \times \mathcal{F} : T_{\max}^\nu(a, f) \leq T\}.$$

If $s < 1/2$, this set is dense in the product of any topology on $\mathcal{X}$ and the relative $L_t^1 H_x^s$ topology on $\mathcal{F}$. Its projection onto $\mathcal{X}$ is all of $\mathcal{X}$. The family obtained while requiring $a = 0$ projects instead to the singleton $\{0\}$.

Proof. For every fixed a , Proposition 3.7 makes the force fiber dense and nonempty. Every product neighborhood of (a, g) therefore meets $\mathfrak{B}_{\nu,T}$ with its first coordinate still equal to a . This proves both density and surjectivity of the projection. Restricting to $a = 0$ gives the last assertion. $\square$

The projection statement has the quantifier order $\forall a \exists f$, not $\exists f \forall a$. For a fixed prescribed force f_* , the set

$$\{a \in \mathcal{X} : T_{\max}^\nu(a, f_*) \leq T\}$$

is a different object and is not classified by these force-changing constructions. There is also a distinction between the two terminal-time assertions. Theorem 3.6 gives singularity exactly at T around every given solution smooth through T . Proposition 3.7 uses breakdown by T , because a given solution may already break down earlier and is then left unchanged. It does not prove that forces producing velocity blowup exactly at T are dense near a force whose solution breaks down earlier.

Remark 3.13. The force convergence in Theorem 3.6 is compatible with diverging pointwise amplitudes. Since the original force is nonzero,

$$\|g_\varepsilon - g\|_{L_{t,x}^\infty} \geq \varepsilon^{-3} \|F\|_\infty - C\varepsilon^{-2} \longrightarrow \infty.$$

Here the first term is the amplitude of F_ε and the second bounds H_ε . The force F is nonzero by (18), since U blows up. Hence the approximating smooth forces have unbounded amplitudes and cannot obey uniform bounds on all derivatives.

3.7 Gluing in a bounded domain

For the following bounded-domain statement, $H^s(\Omega)$ is the space of restrictions of $H^s(\mathbb{R}^3)$ distributions, with quotient norm

$$\|z\|_{H^s(\Omega)} = \inf_{Z|_\Omega = z} \|Z\|_{H^s(\mathbb{R}^3)}. \quad (39)$$

This convention applies also to negative orders. At order zero it is the usual $L^2(\Omega)$ norm. For every fixed compact $K \Subset \Omega$ and $s \in \mathbb{R}$, smooth fields supported in K obey

$$\|z\|_{H^s(\Omega)} \leq \|E_0 z\|_{H^s(\mathbb{R}^3)} \leq C_{s,K,\Omega} \|z\|_{H^s(\Omega)}, \quad \text{supp } z \subset K, \quad (40)$$

Here $E_0 z$ denotes extension by zero. Choose $\chi \in C_c^\infty(\Omega)$ equal to one near K by the smooth Urysohn lemma [9]. Multiplication by a fixed smooth cutoff is bounded on $H^s(\mathbb{R}^3)$ for every real s [13, Chapter 4, Section 2, (2.19)]. For every extension v of z , $\chi v = E_0 z$; taking the infimum over extensions proves the second inequality in (40). The first follows from the quotient norm. The constant depends on the fixed cutoff and s , not on a smaller support or a shrinking scale ε . Applying the pointwise bounds in time gives the same comparison for all the mixed norms in (4), provided K is fixed for all times. No bounded zero-extension operator on arbitrary $H^s(\Omega)$ is asserted. All vector and tensor norms in these definitions are the square root of the sum of the squared component norms.

The next corollary extends the local gluing construction to a bounded container. Placing the modification strictly inside an interior ball leaves the velocity unchanged near the boundary, so the no-slip condition is preserved. The starting point is a given compatible smooth solution; the construction changes its interior evolution while retaining its boundary values.

Corollary 3.14. *Let $\Omega \subset \mathbb{R}^3$ be a bounded box or a bounded smooth domain. Suppose that, for some $\delta > 0$, (v, π, g) is a smooth no-slip solution on $[0, T + \delta]$, and that g is smooth on $\bar{\Omega} \times [0, \infty)$ with temporal support compact in $(0, \infty)$. Here smoothness on a closed spacetime slab means restriction of a C^∞ field from an open neighborhood of that slab; this convention also specifies smoothness at the edges and corners of a box. We assume v, π have this regularity on $\bar{\Omega} \times [0, T + \delta]$, and g on each finite closed slab. The equation and incompressibility hold in Ω , and $v|_{\partial\Omega} = 0$. Pressure may be normalized by zero spatial mean. Existence of this compatible smooth solution is an assumption of the corollary. The construction in Theorem 3.6 applies inside any prescribed interior ball, preserving the initial velocity and no-slip boundary values. The energy estimates use $L^2(\Omega)$ and the force estimates use $L^1(0, \infty; H^s(\Omega))$, with the restriction norm (39). For every real s , the domain and zero-extended force-difference norms are comparable by (40), with a constant independent of ε ; the convergence assertion remains $s < 1/2$.*

Proof. The vector potential and all cutoff estimates are local. Choose the correction and building block supports in a smaller closed ball strictly inside Ω . The computation of the momentum equation in Theorem 3.6 is unchanged. The new velocity equals the given smooth solution in a fixed boundary collar for all times before blowup, so the no-slip boundary values are preserved. The force correction is supported in the same interior region and has the same globally smooth time extension.

For $0 \leq s < 1/2$, the Euclidean scaling proof directly bounds the zero extensions of the force differences in $L^1(0, \infty; H^s(\mathbb{R}^3))$. For $s < 0$, use $\|E_0(g_\varepsilon - g)\|_{H^s(\mathbb{R}^3)} \leq \|E_0(g_\varepsilon - g)\|_{L^2(\mathbb{R}^3)}$ at each time. All these force-difference supports, including those after T , lie in one fixed compact interior ball K for every sufficiently small ε . Consequently (40) gives the asserted domain estimates and the scale-independent comparison after time integration. The energy estimates follow by integrating over that ball. Finally, the classical difference-energy estimate and Grönwall give uniqueness on the no-slip domain; the boundary terms vanish. Thus the constructed solution is singular exactly at T . $\square$

3.8 Further constructions

Proposition 3.15. *Fix the building block (U, P, F) of Theorem 1.1, and fix a spacetime cylinder $Q = B_0 \times (\tau_0, \tau_1)$ with $0 < \tau_0 < \tau_1 < 1$ and $B_0 \Subset \mathbb{R}^3$ an open ball. For any $b \in C_c^\infty(Q; \mathbb{R}^3)$ with $\nabla \cdot b = 0$, define*

$$\begin{aligned}\tilde{U} &= U + b, & \tilde{P} &= P, \\ \tilde{F} &= F + \partial_t b - \nu \Delta b + (U \cdot \nabla)b + (b \cdot \nabla)U + (b \cdot \nabla)b.\end{aligned}\tag{41}$$

These fields give an infinite-dimensional affine family of distinct velocities with zero initial data, finite energy and dissipation, and the same late singular behavior as U . For each fixed nonzero b , the family obtained by replacing b with λb is non-isolated as $\lambda \rightarrow 0$ in every fixed smooth seminorm of the velocity and force differences on their compact support.

Proof. Expanding the momentum equation for $U + b$ gives exactly the force in (41); its divergence remains zero. Every correction term is supported in $\text{supp } b$, a compact subset of Q . The original blowup solution is smooth with bounded derivatives of every fixed order on a neighborhood of that set. Thus the force correction extends smoothly by zero to the whole positive time axis, including time one, despite the singularity of U there.

Since $b = 0$ near time zero and near time one, $\tilde{U}(0) = 0$ and $\tilde{U} = U$ on a fixed late interval. The same unbounded-speed limit holds. The energy and dissipation norms of b are finite by compact smooth support, so the triangle inequality gives the corresponding bounds for $\tilde{U}$.

To verify infinite dimensionality explicitly, choose countably many disjoint small balls inside B_0 . In each ball choose a smooth compactly supported vector potential with nonzero curl, and multiply that curl by a fixed nonzero time bump in (τ_0, τ_1) . These divergence-free fields are linearly independent because their spatial supports are disjoint. Their finite linear combinations belong to the allowed class. The map $b \mapsto U + b$ is injective and affine, so its image is infinite dimensional.

For the non-isolation statement, the velocity difference is λb , while the force difference has the form

$$\lambda L_U b + \lambda^2 (b \cdot \nabla)b, \quad L_U b = \partial_t b - \nu \Delta b + (U \cdot \nabla)b + (b \cdot \nabla)U.$$

On the fixed compact support, all coefficients and their derivatives are bounded. Hence for each integer $m \geq 0$ the C^m norm of the force difference is at most $C_m |\lambda| + C'_m \lambda^2$, and the velocity difference is at most $C''_m |\lambda|$. Both tend to zero. The forces vary in this family; no open singularity basin for one fixed force follows. $\square$

Proposition 3.16. *Fix finitely many disjoint interior balls $B_1, \dots, B_N$ in the torus or in a bounded three-dimensional domain. At fixed $\nu > 0$ and $T > 0$, there is a smooth forced solution with zero initial velocity, smooth on $[0, T)$, with finite energy and dissipation, such that*

$$\limsup_{t \uparrow T} \|u(t)\|_{L^\infty(B_j)} = \infty \quad (1 \leq j \leq N).$$

In a bounded domain its boundary condition is homogeneous no-slip.

Proof. Choose $x_j \in B_j$ and scales $\varepsilon_j > 0$ sufficiently small that the complete scaled compact set $x_j + \varepsilon_j K_*$ lies strictly inside B_j and $\varepsilon_j^2 < T$. Apply (20) to each building block with start time $T - \varepsilon_j^2$, and denote the resulting fields by U_j, P_j, F_j . Set

$$u = \sum_{j=1}^N U_j, \quad p = \sum_{j=1}^N P_j, \quad f = \sum_{j=1}^N F_j.$$

For $i \neq j$, the supports of U_i and U_j are separated, so $(U_i \cdot \nabla)U_j = 0$. The sum therefore satisfies the momentum equation exactly. All force supports are compact in positive time and all forces are smooth. Initial vanishing gives $u(0) = 0$.

On B_j the velocity equals U_j , proving the stated separate limsup for each ball. Disjoint supports also give

$$\sup_{t < T} \|u(t)\|_2^2 \leq M^2 \sum_{j=1}^N \varepsilon_j, \quad \int_0^T \|\nabla u(t)\|_2^2 dt = D^2 \sum_{j=1}^N \varepsilon_j.$$

These sums are finite. In a bounded domain, all fields vanish in a boundary collar; hence $u = 0$ on the boundary. On the torus a spatially constant pressure adjustment gives the chosen gauge. $\square$

Only finitely many solutions are superposed, all with terminal time T . The sequences along which their maxima diverge may differ; the conclusion is a separate limsup statement in each ball.

Proposition 3.17. *On the torus, let $f = -\nabla\phi$ for a globally defined periodic scalar ϕ . On a bounded domain, let $f = -\nabla\phi$ and impose homogeneous no-slip velocity. In either case, any smooth solution with zero initial velocity is identically zero on its classical lifespan.*

Proof. Incompressibility and periodicity or the zero normal boundary trace give

$$\int f \cdot u = - \int \nabla\phi \cdot u = 0.$$

The classical energy identity, integrated from zero, becomes

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = 0.$$

Both terms are nonnegative, so $u(t) = 0$ at every time before blowup. The periodic-potential condition excludes affine potentials, whose gradients need not be removable by the allowed periodic pressure gauge. $\square$

Thus a nontrivial solution with zero initial velocity requires a nonconservative force.

4 The whole space

We now work on $\mathbb{R}^3$, with the data classes $\mathcal{X}_{\mathbb{R}}$ and $\mathcal{F}_{\mathbb{R}}$ from Section 2. The construction is again spatially local. The new issues are the behavior of negative Sobolev norms near frequency zero and the absence of a spectral gap.

4.1 Statement of the density theorem

Theorem 4.1 (Two whole-space Sobolev thresholds). *Fix $\nu, T > 0$, let $q \in \{1, 2\}$, and set $s_q = 2/q - 3/2$. In the relative $L^q(0, \infty; H^s(\mathbb{R}^3))$ topology on $\mathcal{F}_{\mathbb{R}}$:*

- (i) *For every fixed $a \in \mathcal{X}_{\mathbb{R}}$, $\mathcal{B}_{\nu, a, T}^{\mathbb{R}}$ is dense if $s < s_q$.*
- (ii) *For zero initial velocity, $\mathcal{B}_{\nu, 0, T}^{\mathbb{R}}$ is dense if and only if $s < s_q$.*

Thus the thresholds are $1/2$ for $L_t^1 H_x^s$ and $-1/2$ for $L_t^2 H_x^s$. Around every given solution smooth through T , the approximating solution can have the same initial velocity, the same earlier history, singularity exactly at T , and a velocity difference tending to zero in E_T .

The force classes remain smooth, equipped with the relative norm topology. For $q = 2$, the converse uses a regular neighborhood whose radius depends on T .

4.2 Local construction and negative Sobolev estimates

The compact solution of Theorem 1.1 is already defined on $\mathbb{R}^3$. Use the scaling (20) directly, without periodization. The energy identities (21) and mixed force identity (22) follow from the same changes of variables. The vector potential in Lemma 3.4 is local, so its proof applies in any Euclidean ball. In Lemma 3.5, the support, derivative, energy, and mixed Lebesgue bounds are also local; only the final transfer of fractional norms to the torus is unnecessary. Thus the same w_ε and H_ε are available on $\mathbb{R}^3$, with the same uniform bounds. We record the result and prove the additional negative-order estimates.

Theorem 4.2. *Let (v, π, g) be the solution for $a \in \mathcal{X}_\mathbb{R}$ and $g \in \mathcal{F}_\mathbb{R}$, smooth through $T + \delta$ for some $\delta > 0$. Fix any nonempty open ball $B \subset \mathbb{R}^3$. For all sufficiently small $\varepsilon > 0$, there are $g_\varepsilon \in \mathcal{F}_\mathbb{R}$ and a solution u_ε such that*

$$T_{\max, \mathbb{R}}^\nu(a, g_\varepsilon) = T, \quad \limsup_{t \uparrow T} \|u_\varepsilon(t)\|_\infty = \infty, \quad u_\varepsilon = v \quad (0 \leq t \leq T - 2\varepsilon^2).$$

The velocity difference is supported inside B at every $t < T$, and $g_\varepsilon - g \in C_c^\infty(B \times (0, \infty))$. Moreover,

$$\|u_\varepsilon - v\|_{E_T} \leq (M + D)\varepsilon^{1/2} + C\varepsilon^{3/2}. \quad (42)$$

For $q \in \{1, 2\}$ the force difference tends to zero in $L_t^q H_x^s$ whenever $s < 2/q - 3/2$. All of these conclusions hold for the same family of glued solutions.

Proof. Place the fields in (20) inside B , and choose the compact cutoffs of Lemma 3.4. Define $A, w_\varepsilon, H_\varepsilon$ by (24), (25) and (27). These formulas only use v on a fixed compact neighborhood of B and of time T , so the preceding lemmas apply. In particular $v + w_\varepsilon = 0$ on a neighborhood of the active building block support. Put

$$u_\varepsilon = v + w_\varepsilon + U_\varepsilon, \quad p_\varepsilon = \pi + P_\varepsilon, \quad g_\varepsilon = g + H_\varepsilon + F_\varepsilon.$$

Writing $b_\varepsilon = v + w_\varepsilon$, expansion of the equation leaves only the cross-advection terms $(b_\varepsilon \cdot \nabla)U_\varepsilon + (U_\varepsilon \cdot \nabla)b_\varepsilon$. They vanish on a neighborhood of the building block support because $b_\varepsilon = 0$ there. Outside that support, the building block and its derivatives vanish; smoothness gives the same conclusion at the support boundary. When the building block is inactive it is identically zero. Thus the two terms vanish pointwise everywhere, and the equation is exact. The pressure difference may be chosen to be the compact scalar P_ε : the equation gives an L^2 pressure gradient, and applying $I - \mathbb{P}$ recovers (15). Both force corrections are globally smooth and spacetime compact, including across T , so their sum preserves membership in $\mathcal{F}_\mathbb{R}$.

The initial value and earlier history are unchanged. At each $t < T$ the compact blowup solution and cutoff correction lie in H^∞ , and the maximum norm of U_ε is unbounded at T . Proposition 2.1 identifies the solution with the unique maximal solution. An extension through T would be bounded in $C_t H^2$ on a neighborhood of T , hence bounded in L_x^∞ by (12), contradicting the blowup of U_ε . Thus its maximal lifespan is exactly T . Equation (21) and the correction estimate in Lemma 3.5 give (42) by the triangle inequality.

It remains to justify every asserted force topology, including negative orders. Let

$$\beta(q, s) = \frac{2}{q} - \frac{3}{2} - s.$$

For $0 \leq s \leq 1$, Euclidean scaling and $\|z\|_{H^s} \leq C_s(\|z\|_2 + \|z\|_{\dot{H}^s})$ give

$$\begin{aligned} \|F_\varepsilon\|_{L_t^q H_x^s} &\leq C_{q,s}(\varepsilon^{2/q-3/2} + \varepsilon^{\beta(q,s)}), \\ \|H_\varepsilon\|_{L_t^q H_x^s} &\leq C_{q,s}(\varepsilon^{2/q-1/2} + \varepsilon^{\beta(q,s)+1}). \end{aligned} \quad (43)$$

The second line uses the uniformly smooth compact rescaled profile of Lemma 3.5, whose amplitude is ε^{-2} .

For $-3/2 < s < 0$, every smooth compact profile has finite $\dot{H}^s$ norm. Indeed, on $|\xi| < 1$ its Fourier transform is bounded by its L^1 norm and $\int_{|\xi| < 1} |\xi|^{2s} d\xi < \infty$; on $|\xi| \geq 1$ its L^2 norm suffices. These bounds are uniform for the rescaled correction profiles, which have common compact support and uniform derivatives. Since $(1 + |\xi|^2)^s \leq |\xi|^{2s}$, scaling gives

$$\|F_\varepsilon\|_{L_t^q H_x^s} \leq C_{q,s} \varepsilon^{\beta(q,s)}, \quad \|H_\varepsilon\|_{L_t^q H_x^s} \leq C_{q,s} \varepsilon^{\beta(q,s)+1}. \quad (44)$$

The time norms run over the entire positive axis. Each rescaled force has time support of length $O(\varepsilon^2)$, including any part after T .

For $q = 1$, (43) proves convergence for $0 \leq s < 1/2$; all $s < 0$ follow from $\|z\|_{H^s} \leq \|z\|_2$. For $q = 2$, (44) proves convergence for $-3/2 < s < -1/2$. If $s \leq -3/2$, choose $r \in (-3/2, -1/2)$ with $r > s$ and use $\|z\|_{H^s} \leq \|z\|_{H^r}$. This last step avoids making any false homogeneous scaling assertion at indices where a generic compact profile can have an infinite homogeneous norm. $\square$

4.3 The critical estimate for time-integrable forcing

Proposition 4.3. *There is a universal $c > 0$ such that, for $a \in \mathcal{X}_{\mathbb{R}}$ and $f \in \mathcal{F}_{\mathbb{R}}$,*

$$\|a\|_{\dot{H}^{1/2}} + \|f\|_{L_t^1 \dot{H}_x^{1/2}} < c\nu \implies T_{\max, \mathbb{R}}^\nu(a, f) = \infty.$$

In particular, for $a = 0$, the ball $\|f\|_{L_t^1 \dot{H}_x^{1/2}} < c\nu$ consists of globally regular inputs.

Proof. This is the $p = r = 2$ case of Danchin's forced critical small-data theorem [15, Theorem 2.3.1, pp. 47–49]. Indeed, $\dot{B}_{2,2}^{1/2} = \dot{H}^{1/2}$ and $\|f\|_{\tilde{L}^1 \dot{H}^{1/2}} \leq \|f\|_{L^1 \dot{H}^{1/2}}$; the Leray projection is bounded on these spaces. The theorem gives a global critical solution, and persistence of regularity for the smooth data in our classes identifies it with the maximal classical solution of Proposition 2.1. Finally $\|f\|_{\dot{H}^{1/2}} \leq \|f\|_{H^{1/2}}$ gives the stated open ball. $\square$

4.4 The critical estimate for square-integrable forcing

Proposition 4.4. *For each $\nu, S > 0$ there is $r_{\nu,S} > 0$ such that*

$$f \in \mathcal{F}_{\mathbb{R}}, \quad \|f\|_{L^2(0,\infty; H^{-1/2})} < r_{\nu,S} \implies T_{\max, \mathbb{R}}^\nu(0, f) > S.$$

One can take $r_{\nu,S} = c\nu^{3/2}e^{-C\nu S}$ for suitable universal positive constants c, C .

Proof. Use the inhomogeneous heat estimate [15, Theorem 2.2.5, p. 44] and the critical bilinear fixed-point argument [15, Lemma 2.3.2 and the proof of Theorem 2.3.1, pp. 48–49]. We spell out the low-frequency and viscosity adjustments. Normalize viscosity by $u(t, x) = \nu v(\nu t, x)$ and $F(\tau, x) = \nu^{-2} f(\tau/\nu, x)$. On $0 \leq \tau \leq L := \nu S$, set

$$X_L = C([0, L]; H^{1/2}) \cap L^2(0, L; H^{3/2}), \quad \mathcal{H}F(t) = \int_0^t e^{(t-r)\Delta} \mathbb{P}F(r) dr.$$

The cited heat estimate and the Sobolev product law give, with the sum norm on X_L ,

$$\|\mathcal{H}F\|_{X_L} \leq Ce^{CL} \|F\|_{L^2 H^{-1/2}}, \quad \|\mathcal{B}(v, w)\|_{X_L} \leq Ce^{CL} \|v\|_{X_L} \|w\|_{X_L},$$

where $\mathcal{B}(v, w) = -\mathcal{H} \operatorname{div}(v \otimes w)$. For the second bound use $X_L \hookrightarrow L_t^4 H_x^1$ and $H^1 \cdot H^1 \hookrightarrow H^{1/2}$; the exponential is a convenient upper bound for the finite-time constants in the inhomogeneous heat estimate. Thus the cited contraction lemma applies if $\|F\|_{L^2 H^{-1/2}} < ce^{-CL}$, after changing universal constants. Since $\|F\|_{L^2(0,L; H^{-1/2})} = \nu^{-3/2} \|f\|_{L^2(0,S; H^{-1/2})}$, the stated radius suffices. Persistence of smoothness and local existence at the final time give lifespan strictly greater than S . The finite-time inhomogeneous estimate is essential: small $H^{-1/2}$ norm does not bound the homogeneous negative norm at low frequencies. $\square$

Proof of Theorem 4.1. Fix a, g and a positive radius in the indicated force norm. There are two cases. If $T_{\max, \mathbb{R}}^\nu(a, g) \leq T$, take $f = g$. Otherwise choose $\delta > 0$ so that the given smooth solution exists through $T + \delta$, and apply Theorem 4.2. For $s < s_q$, its force difference is smaller than the prescribed radius when ε is sufficiently small, its initial velocity is exactly a , and its lifespan is exactly T . This proves density for each fixed smooth initial velocity.

For zero initial velocity and $q = 1$, Proposition 4.3 provides an open ball about zero disjoint from $\mathcal{B}_{\nu, 0, T}^\mathbb{R}$ in $L_t^1 H_x^{1/2}$. The inclusion $H^s \hookrightarrow H^{1/2}$ with norm at most one for $s \geq 1/2$ gives the same obstruction in each stronger metric. For $q = 2$, use Proposition 4.4 with $S = T$ and then $H^s \hookrightarrow H^{-1/2}$ for $s \geq -1/2$. These nonempty relative open balls prove non-density. The energy and earlier-history assertions are part of the gluing theorem. $\square$

4.5 Compact support, rapid decay, and completed force spaces

Define the following two subclasses of $\mathcal{F}_\mathbb{R}$:

$$\mathcal{F}_c = C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3),$$

$$\mathcal{F}_{\text{rd}} = \left\{ f \in C^\infty(\mathbb{R}^3 \times [0, \infty)) : \sup_{x \in \mathbb{R}^3, t \geq 0} (1 + |x| + t)^N |\partial_x^\alpha \partial_t^j f(x, t)| < \infty \text{ for all } N, \alpha, j \right\}.$$

Vector values are understood in the second line. The initial class corresponding to the rapid-decay formulation is $\mathcal{S}_\sigma = \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3) \cap L_\sigma^2$. These are exactly the decay requirements relevant to [3, equations (4)–(5)]; a common numerical bound on their seminorms is not imposed.

Corollary 4.5. *Theorem 4.1 remains valid with $\mathcal{F}_\mathbb{R}$ replaced by $\mathcal{F}_c$ or $\mathcal{F}_{\text{rd}}$. In particular it holds for every fixed $a \in \mathcal{S}_\sigma$ in the rapid-decay class, with the complete if-and-only-if classification at $a = 0$.*

Proof. The initial value is preserved exactly, and the force correction in Theorem 4.2 is spacetime compact. Addition of that correction preserves each stated force class, including all rapid-decay bounds. The two-case density proof therefore stays within the chosen subclass. Each critical regular ball contains zero and remains a nonempty relative open ball in that subclass. The local framework only requires $a \in \mathcal{X}_\mathbb{R}$, so it applies in particular to Schwartz initial data. Nothing here requires the nonzero given velocity or its pressure to be spatially compact. $\square$

The force spaces in the next proposition are natural for the energy class. With $V = H^1(\mathbb{R}^3; \mathbb{R}^3) \cap L_\sigma^2$, a finite-energy velocity belongs to $L_t^\infty L_\sigma^2 \cap L_t^2 V$. A full distribution in H^{-1} restricts to an element of V' ; the latter records only its action on divergence-free tests. This distinction is discussed in [1, Definition 1 and Remark 4]. Our realization of $\dot{H}^{-1}$ gives the pairing

$$|\langle h, z \rangle| \leq \|h\|_{\dot{H}^{-1}} \|\nabla z\|_2, \quad \left| \int_0^T \langle f, u \rangle \, dt \right| \leq \|f\|_{L_t^2 \dot{H}_x^{-1}} \|\nabla u\|_{L_{t,x}^2}.$$

These inequalities follow by Fourier Cauchy–Schwarz, first for Schwartz functions and then by completion. The inhomogeneous H^{-1} norm instead pairs with H^1 , while $L_t^1 L_x^2$ pairs with $L_t^\infty L_x^2$.

Proposition 4.6. *Fix $a \in \mathcal{X}_\mathbb{R}$ and $\nu, T > 0$. Smooth compact forces with $T_{\max, \mathbb{R}}^\nu(a, f) \leq T$ are dense in each full Bochner space $L^q(0, \infty; H^s(\mathbb{R}^3))$ for $q \in \{1, 2\}$ and $s < s_q$. They are also dense in $L^2(0, \infty; \dot{H}^{-1}(\mathbb{R}^3))$.*

For every given smooth solution in Theorem 4.2, one may simultaneously arrange

$$\|u_\varepsilon - v\|_{E_T} \longrightarrow 0, \quad \|g_\varepsilon - g\|_{L_t^1 L_x^2} + \|g_\varepsilon - g\|_{L_t^2 H_x^{-1}} + \|g_\varepsilon - g\|_{L_t^2 \dot{H}_x^{-1}} \longrightarrow 0.$$

The homogeneous norm here is a norm of the compact difference; the given force itself need not belong to that homogeneous force space.

Proof. The standard density of Schwartz functions in H^s [13, Chapter 4, Section 1, Exercise 1], followed by smooth spatial cutoffs, gives density of C_c^∞ for every real s . For the homogeneous space, approximate $|\xi|^{-1}\hat{h}$ by smooth annular Fourier data and then use the same cutoffs. The only additional low-frequency estimate is

$$\|k\|_{\dot{H}^{-1}}^2 \leq C \|k\|_1^2 \int_{|\xi|<1} |\xi|^{-2} d\xi + \|k\|_2^2 \leq C' \|k\|_1^2 + \|k\|_2^2. \quad (45)$$

Applied to the Schwartz cutoff tails, this proves spatial compact-smooth density in $\dot{H}^{-1}$. The constructions preserve real vector fields. For time dependence, Hunter [6, Proposition 6.29] gives density of finite sums $\sum_j \varphi_j(t)b_j$ with $\varphi_j \in C_c^\infty$ in $L^q(I; X)$, $q < \infty$. Truncate $(0, \infty)$ to $[1/N, N]$ and approximate each b_j by a spatially compact smooth field. The resulting finite sums are jointly smooth and compactly supported in space and positive time.

For the inhomogeneous density of forces producing breakdown, first choose such a smooth compact force within half the prescribed radius of the target. Corollary 4.5 supplies a smooth compact force producing breakdown within the other half. This is density of a smooth subset in the completion, not a definition of classical breakdown for every rough force.

For the homogeneous gluing estimate, apply the Fourier scaling calculation of (44) at $s = -1$. Explicitly, an amplitude ε^{-a} has spatial $\dot{H}^{-1}$ factor $\varepsilon^{5/2-a}$, because $\widehat{\varepsilon^{-a}h(\cdot/\varepsilon)}(\xi) = \varepsilon^{3-a}\hat{h}(\varepsilon\xi)$. The time L^2 factor is ε . The source force has $a = 3$, while the correction has $a = 2$ and uniformly bounded compact rescaled profiles; their $\dot{H}^{-1}$ norms are uniformly finite by (45). Hence

$$\|F_\varepsilon\|_{L_t^2 \dot{H}_x^{-1}} \leq C\varepsilon^{1/2}, \quad \|H_\varepsilon\|_{L_t^2 \dot{H}_x^{-1}} \leq C\varepsilon^{3/2}.$$

Given a compact smooth force, if its lifespan is at most T it already belongs to the required breakdown set. Otherwise glue the building block and use these estimates. Combining this relative approximation with the proved compact-smooth Bochner density establishes density in $L_t^2 \dot{H}_x^{-1}$. Finally these estimates, (42), and the $q = 1, s = 0$ and $q = 2, s = -1$ cases of the gluing estimates prove the simultaneous convergence for the same family. $\square$

As on the torus, every $a \in \mathcal{X}_\mathbb{R}$ occurs as the initial velocity of a solution producing breakdown for some smooth force. Thus the projection of the initial data and forces producing breakdown onto $\mathcal{X}_\mathbb{R}$ is the whole space, with quantifier order $\forall a \exists f$. The preceding approximation concerns finite-energy whole-space solutions and uses one rescaled copy of U in each perturbation.

4.6 Cell averages on prescribed grids

The spaces used for energy estimates should be distinguished from the additional regularity assumed in numerical error estimates. For example, on a bounded no-slip computational domain one uses the solenoidal energy space H with zero normal trace and $V = H_0^1 \cap H$; finite-dimensional velocity and pressure spaces discretize this setting. Guermond, Mineev and Shen [4, Section 2 and Theorem 3.2] state these spaces and the additional smoothness required for their error estimate. The initial pressure is constrained by the equation and boundary conditions. These bounded-domain conventions do not impose boundary conditions at infinity in the whole-space result below.

To relate the continuous norms to a concrete numerical observation, let $\mathcal{T}_h$ be a complete uniform Cartesian grid of $\mathbb{R}^3$ with positive mesh widths. For a locally integrable vector field, define the cell-observation map with sequence codomain

$$A_h : L_{\text{loc}}^1(\mathbb{R}^3; \mathbb{R}^3) \longrightarrow (\mathbb{R}^3)^{\mathcal{T}_h}, \quad (A_h z)_C = \frac{1}{|C|} \int_C z(x) dx.$$

Only coordinatewise equality in this codomain is needed. If a sequence norm is desired for $z \in L^2$, the natural volume-weighted norm obeys $\sum_{C \in \mathcal{T}_h} |C| |(A_h z)_C|^2 \leq \|z\|_2^2$ by Cauchy–Schwarz in

each cell and summation. Each grid has infinitely many cells. The next result concerns a finite number of such grids and equality on every cell of each grid, at every time before blowup.

Theorem 4.7. *Under the hypotheses on the given smooth solution of Theorem 4.2, fix a finite family of complete uniform Cartesian grids. The glued solutions may be chosen so that, for every grid in that family,*

$$A_h u_\varepsilon(t) = A_h v(t), \quad A_h g_\varepsilon(t) = A_h g(t) \quad (0 \leq t < T),$$

while $T_{\max, \mathbb{R}}^\nu(a, g_\varepsilon) = T$ and the energy and force convergences in Proposition 4.6 hold. All differences are supported inside a ball contained in one cell of every grid, except for an optional spatially constant pressure gauge.

Proof. The union of faces in one grid is a closed, locally finite union of planes. For the fixed finite family their union is closed and has measure zero. Choose a point outside that union and a ball with closure inside one cell of each grid. Apply Theorem 4.2 inside that ball, using the compact pressure representative. Put $\delta u = u_\varepsilon - v$, $\delta p = p_\varepsilon - \pi = P_\varepsilon$, and $\delta g = g_\varepsilon - g$.

Fix one containing cell C . For every $t < T$, δu is smooth, divergence free and compactly supported in its interior. For component j ,

$$\delta u_j = \nabla \cdot (x_j \delta u), \quad \int_C \delta u_j \, dx = \int_{\partial C} x_j \delta u \cdot n \, dS = 0.$$

Subtract the two momentum equations and integrate to obtain

$$\int_C \delta g \, dx = \frac{d}{dt} \int_C \delta u \, dx + \int_{\partial C} (u_\varepsilon \otimes u_\varepsilon - v \otimes v + \delta p I - \nu \nabla \delta u) n \, dS. \quad (46)$$

The time derivative is zero by the preceding identity. All differences vanish in a neighborhood of the cell boundary, so the surface term is zero. A spatially constant pressure gauge adds a multiple of $\int_{\partial C} n \, dS = 0$. The averages therefore agree in C ; in every other cell they agree by support. The same construction fits the containing cell of each grid, and the convergence estimates are unaffected by this fixed upper bound on ε . $\square$

A deterministic procedure using only the initial velocity cell averages and the force cell averages for $0 \leq t < T$ receives identical data in the two cases. Other observations, such as point values or pressure, may distinguish them. The perturbation depends on the fixed grid family and develops larger amplitudes and derivatives as its support shrinks. The result therefore concerns the information in the prescribed averages, rather than convergence under refinement for one fixed smooth problem.

Declaration of generative AI use

GPT Astra (OpenAI) assisted with literature review, development and checking of mathematical arguments, drafting, and LaTeX preparation. Codex assisted with manuscript consolidation and revision. Responsibility for the final content remains with the authors.

References

- [1] L. C. Berselli and S. Spirito. On the existence of Leray–Hopf weak solutions to the Navier–Stokes equations. *Fluids* **6**(1) (2021), 42. doi:10.3390/fluids6010042.
- [2] A. Enciso, J. Peñafiel-Tomás and D. Peralta-Salas. An extension theorem for weak solutions of the 3d incompressible Euler equations and applications to singular flows. *Forum of Mathematics, Pi* **13** (2025), e21. doi:10.1017/fmp.2025.10012.

- [3] C. L. Fefferman. Existence and smoothness of the Navier–Stokes equation. In J. Carlson, A. Jaffe and A. Wiles (eds.), *The Millennium Prize Problems*. Clay Mathematics Institute and American Mathematical Society, 2006, pp. 57–67.
- [4] J. L. Guermond, P. Mineev and J. Shen. An overview of projection methods for incompressible flows. *Computer Methods in Applied Mechanics and Engineering* **195**(44–47) (2006), 6011–6045. doi:10.1016/j.cma.2005.10.010.
- [5] M. Hofmanová, R. Zhu and X. Zhu. Non-uniqueness of Leray–Hopf solutions for stochastic forced Navier–Stokes equations. arXiv:2309.03668v2, 2024.
- [6] J. K. Hunter. *Notes on Partial Differential Equations*. University of California, Davis, 2014. https://www.math.ucdavis.edu/~hunter/pdes/pde_notes.pdf.
- [7] OpenAI. *Finite Time Blowup for Navier–Stokes*. 2026. <https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf>.
- [8] OpenAI. *NavierStokesAndEuler*. Software, 2026. <https://github.com/openai/NavierStokesAndEuler>.
- [9] P. Petersen. *Manifolds, Transversality, and de Rham Cohomology*. University of California, Los Angeles, lecture notes. <https://www.math.ucla.edu/~petersen/manifolds.pdf>.
- [10] T. Tao. *Nonlinear Dispersive Equations: Local and Global Analysis*. CBMS Regional Conference Series in Mathematics, vol. 106. American Mathematical Society, 2006. doi:10.1090/cbms/106.
- [11] T. Tao. Localisation and compactness properties of the Navier–Stokes global regularity problem. *Analysis & PDE* **6**(1) (2013), 25–107. doi:10.2140/apde.2013.6.25.
- [12] T. Tao. 254A, Notes 1: Local well-posedness of the Navier–Stokes equations. *What’s New*, 16 September 2018. <https://terrytao.wordpress.com/2018/09/16/254a-notes-1-local-well-posedness-of-the-navier-stokes-equations/>.
- [13] M. E. Taylor. *Partial Differential Equations I: Basic Theory*. 2nd ed., Applied Mathematical Sciences, vol. 115. Springer, 2011. doi:10.1007/978-1-4419-7055-8.
- [14] M. E. Taylor. *Partial Differential Equations III: Nonlinear Equations*. 2nd ed., Applied Mathematical Sciences, vol. 117. Springer, 2011. doi:10.1007/978-1-4419-7049-7.
- [15] R. Danchin. *Fourier Analysis Methods for PDE’s*. Lecture notes, 2005. <https://perso.math.u-pem.fr/danchin.rafael/cours/courschaine.pdf>.
- [16] P. Marín-Rubio, J. C. Robinson and W. Sadowski. Solutions of the 3D Navier–Stokes equations for initial data in $H^{1/2}$: Robustness of regularity and numerical verification of regularity for bounded sets of initial data in $\dot{H}^1$. *Journal of Mathematical Analysis and Applications* **400**(1) (2013), 76–85. doi:10.1016/j.jmaa.2012.10.064.